

CardioVision AI: A Multimodal Vision Transformer Framework for Cardiovascular Event Risk Assessment

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Abstract—Coronary artery disease is among the foremost contributors to cardiovascular mortality worldwide, creating a pressing need for reliable automated tools that go beyond simple image interpretation. This paper introduces CardioVision AI, a multimodal deep learning framework that jointly analyses coronary angiography and patient-level clinical data to classify stenosis severity and estimate cardiovascular event risk. A Vision Transformer backbone processes individual angiographic frames from the CADICA dataset, while a four-layer clinical multi-layer perceptron encodes population-scale cardiovascular risk features drawn from the NHANES survey. A compact learned FusionGate dynamically weights the two branches at inference time, producing a three-class output covering Normal, Mild, and Severe stenosis. Across a six-variant ablation study, the proposed architecture attains 94.22% accuracy and a macro F1 score of 0.932, outperforming both a convolutional baseline and a ResNet-based fusion model by substantial margins. Prediction transparency is provided through a blended Grad-CAM++ and SmoothGrad saliency pipeline that highlights the vessel regions driving each classification. A standalone 10-year cardiovascular risk estimator grounded in the 2013 ACC/AHA Pooled Cohort Equations complements the imaging output. All components are integrated into a four-tab interactive Streamlit application for clinical and academic review.

Index Terms—coronary artery disease, vision transformer, multimodal fusion, stenosis classification, NHANES, CADICA, Grad-CAM++, ASCVD risk, deep learning, clinical decision support

I. INTRODUCTION

CORONARY artery disease accounts for approximately 17.9 million deaths each year, making it the single largest cause of mortality across the globe [1]. Its diagnosis depends heavily on invasive coronary angiography, in which a contrast agent is injected into the arterial tree and a trained specialist visually estimates the degree to which each vessel is narrowed.

This assessment is inherently subjective, varies considerably between observers, and requires a level of subspecialty expertise that is seldom available outside well-resourced tertiary hospitals.

Deep learning has substantially advanced automated analysis in medical imaging, and convolutional networks have been applied with growing success to angiogram interpretation over the past decade [2]–[5]. A persistent weakness in nearly all of these systems, however, is that they treat the angiographic image as a self-contained signal. In clinical practice, a cardiologist reads the angiogram against everything already known about the patient. Consider the contrast between a 72-year-old male smoker with poorly controlled diabetes and an LDL of 280 mg/dL, and a 38-year-old non-smoking woman presenting with an identical angiographic appearance. Their underlying risk profiles differ profoundly, yet most automated systems would assign them the same prediction. Routine laboratory measurements, lifestyle indicators, and demographic factors capture exactly this context, yet they are almost universally absent from automated angiography pipelines.

A further practical obstacle is the lack of a publicly available, sufficiently large angiography dataset that also contains matched patient-level clinical metadata. CardioVision AI addresses this gap by strategically combining two independent public sources: the CADICA coronary angiography dataset [6] and the National Health and Nutrition Examination Survey [7]. Because these datasets contain entirely different individuals, patient-level record linkage is neither feasible nor necessary. The proposed framework bridges them instead through *shared clinical semantics*: an XGBoost classifier trained on NHANES learns to map cardiovascular risk factors onto the same three-level severity scale used in CADICA, generating class-conditional clinical representations that reflect what a high-

risk, moderate-risk, or low-risk patient looks like in terms of blood chemistry and lifestyle. No NHANES respondent has ever appeared in a CADICA angiogram, yet the representations transfer meaningfully to individual patients at deployment time.

This approach of fusing information from non-overlapping cohorts through learned feature representations rather than shared patient identifiers is well established in multimodal medical AI. Acosta et al. [21] demonstrated that retinal photographs and electronic health records drawn from entirely different patient groups can be merged at the feature level to generate cardiovascular risk estimates, without requiring a single patient to appear in both data sources. The same principle governs the present work: two independently trained branches, each estimating the same three-class outcome from a different modality, are combined at the logit level so that at test time both branches operate on data from the same individual, producing a genuinely patient-specific result.

From the image modelling perspective, Vision Transformers offer a structural advantage over convolutional networks for reading coronary angiograms. Identifying a stenosis requires spatial reasoning that extends well beyond the immediately surrounding pixels—the narrowing must be understood in relation to the broader vessel tree, the contrast wash-through pattern, and overall coronary anatomy. The global self-attention mechanism in ViT-Base/16 is architecturally well suited to this kind of relational reasoning, whereas convolutions, regardless of depth, are limited to features that can be built from progressively expanded local neighbourhoods.

The primary contributions of this work are as follows:

- A cross-dataset multimodal fusion architecture coupling a ViT-Base/16 image branch with a four-layer clinical MLP, combined through a compact learned FusionGate for dynamic per-sample modality weighting.
- A class-conditional clinical prior construction strategy that bridges population-level NHANES data with frame-level CADICA labels without requiring any shared patient identifiers.
- A blended Grad-CAM++ and SmoothGrad explainability pipeline delivering class-discriminative saliency maps with a three-level bounding box fallback that guarantees a localisation region on every input image.
- A post-training logit debiasing scheme that corrects the minority-class amplification induced by Focal Loss training without requiring the model to be retrained or recalibrated.
- A standalone validated implementation of the 2013 ACC/AHA Pooled Cohort Equations providing complementary 10-year cardiovascular event probability estimates.
- A structured six-variant ablation study with bootstrap confidence intervals and McNemar significance testing to isolate the contribution of each architectural component.
- A four-tab Streamlit application that exposes all components for clinical and academic evaluation.

II. RELATED WORK

A. Deep Learning for Coronary Angiography

Early efforts at automating coronary lesion detection centred on adapting widely used convolutional architectures such as VGG and ResNet to individual frames extracted from clinical angiographic sequences [4]. Recognising that single frames discard temporal context, later work processed full video clips in an attempt to capture the dynamics of contrast agent propagation through the vessel tree [5]. Both lines of work highlighted a common bottleneck: the absence of a large, publicly available dataset with standardised stenosis severity annotations. The CADICA collection was assembled to fill exactly that gap, providing frame-level labels annotated by clinical experts from a real patient population [6]. This work adopts CADICA as its imaging foundation, remapping its original per-frame stenosis percentage categories to the three clinically grounded severity classes defined in Section III.

B. Vision Transformers in Medical Imaging

The original ViT publication established that a purely attention-based architecture could match convolutional models on large-scale natural image classification when pre-trained on sufficiently large corpora [8]. This finding prompted rapid exploration in medical imaging, where tasks such as pathology segmentation and retinal lesion detection share the property that relevant information is spread across the image rather than concentrated in a single local region [10]. The global receptive field provided by self-attention is a natural match for this class of problem. Coronary angiography shares the same characteristic, and the present study is one of the first to evaluate ViT-Base/16 on the CADICA stenosis grading benchmark.

C. Multimodal Fusion in Clinical AI

The fusion of image features with structured clinical measurements has been explored through a progression of increasingly flexible architectures. The simplest strategy concatenates the two feature vectors upstream of a shared classifier [11]. Learnable attention-based gating mechanisms, which assign modality-specific weights dynamically depending on input quality, offer greater adaptability [13]. Co-attention transformers that model bidirectional interactions between paired modalities have been proposed for survival prediction from histopathology whole-slide images [12]. The FusionGate introduced here occupies a middle ground: it operates directly on the three-class logit outputs of each branch rather than on high-dimensional feature embeddings, learning per-class weighting scalars that make the modality preference interpretable and the module lightweight at just 224 trainable parameters.

D. Cross-Dataset and Cross-Cohort Fusion

Fusing information across datasets with non-overlapping patient populations remains a considerably less explored problem than within-cohort multimodal learning. Acosta et al. [21] provided direct evidence that cardiovascular event risk can be predicted by combining retinal images and clinical

records drawn from entirely separate cohorts, without any requirement for shared patient identifiers between the imaging and clinical sources. A systematic review by Stahlschmidt et al. [22] further confirmed that cross-cohort feature-level fusion is methodologically defensible whenever both modalities are informative about the same clinical target. The present work applies this framework specifically to the pairing of coronary angiographic frames and national population health surveys, which to our knowledge has not previously been attempted.

E. Explainability in Cardiovascular AI

Grad-CAM [14] generates class-discriminative saliency maps by projecting gradient information, weighted by class-specific activations, back onto intermediate feature layers. Grad-CAM++ [15] extends this formulation with second-order gradient weighting, which concentrates the highlighted region on the most influential spatial locations rather than spreading activation broadly—an important property when the target is a localised vascular narrowing. SmoothGrad [16] reduces the pixel-level noise inherent in single-pass gradient attribution by averaging attributions over many slightly perturbed input realisations. Attention rollout [17] offers an alternative by propagating attention weights through all transformer layers, but because it operates without class specificity, it tends to highlight the entire visible vasculature uniformly regardless of what the model predicts. For this reason, the present pipeline relies on the two class-specific methods rather than rollout.

III. DATASETS AND PREPROCESSING

A. CADICA Angiography Dataset

CADICA [6] is an open-access benchmark of coronary angiography frames with per-frame stenosis percentage labels assigned by expert clinical annotators. For this work, the original categorical labels are remapped to three clinically meaningful severity classes according to the scheme shown in Table I. Frames without any haemodynamically relevant narrowing are assigned to the Normal class. Those in the moderate stenosis range form the Mild class. Frames at or above 60% narrowing—the widely accepted threshold at which revascularisation becomes clinically indicated—are placed in the Severe class.

TABLE I
CADICA RAW CATEGORY TO THREE-CLASS LABEL MAPPING

CADICA Raw Category	Class	Index
p0, p0_20	Normal	0
p20, p20_35, p35, p35_60, p50, p0_50, p20_50	Mild	1
p60, p60_100, p50_70, p50_100, p70, p70_100, p100	Severe	2

The full dataset totals 22,761 frames and displays pronounced class imbalance, with Normal frames substantially outnumbering Severe ones. To guard against data leakage, splitting is carried out at the patient level rather than the frame level, so all frames from a single patient are confined to one partition. The resulting allocation is 70% of patients for training, 15% for validation, and 15% for testing. All frames

are resized to 224×224 pixels and normalised using ImageNet channel statistics (mean [0.485, 0.456, 0.406], standard deviation [0.229, 0.224, 0.225]).

B. Data Augmentation

Training applies a set of online augmentations chosen specifically to introduce variation that mirrors real acquisition variability without altering diagnostically relevant content. Random horizontal flipping (probability 0.5), random rotation within ± 10 degrees, and mild colour jitter affecting brightness and contrast (± 0.1) are applied on-the-fly during each training epoch. Validation and test images are processed without augmentation to ensure that reported metrics reflect true generalisation.

C. NHANES Clinical Dataset

The National Health and Nutrition Examination Survey [7] is a nationally representative cross-sectional study covering 11,933 adult participants. Features used in this work span four domains. *Laboratory measurements* include total cholesterol, HDL cholesterol, triglycerides, fasting glucose, HbA1c, high-sensitivity C-reactive protein (hs-CRP), and fasting insulin. *Blood pressure and medication* fields cover systolic blood pressure, antihypertensive medication use, and lipid-lowering medication use. *Lifestyle and dietary* variables include vigorous physical activity participation, sedentary hours per day, saturated fat intake, sodium intake, dietary cholesterol intake, and home-prepared meal frequency. *Derived and demographic* fields encompass BMI, HOMA-IR, non-HDL cholesterol, triglyceride-to-HDL ratio, smoking status, diabetes and hypertension binary flags, age, sex, and ethnicity encoded from the RIDRETH3 variable.

All missing values are imputed with column-level medians. Features are standardised using a StandardScaler fitted exclusively on the NHANES training subset and saved for use during deployment. The dimensionality of the resulting feature vector is resolved automatically at runtime from the saved scaler, so changes to the NHANES feature set propagate through the inference pipeline without requiring any code modifications.

D. Cross-Dataset Clinical Prior: Bridging CADICA and NHANES

The central methodological challenge is to combine training data from two datasets whose patients share no common identifiers. The solution is to operate at the level of class-conditional feature distributions rather than at the level of individual records.

An XGBoost classifier is trained on the NHANES cohort to predict cardiovascular risk category from the clinical features described above. Its output classes are intentionally aligned with the three CADICA severity labels. NHANES participants are then partitioned into three class-specific pools according to their predicted label. During training, each CADICA frame with ground-truth severity label k is paired with a clinical feature vector sampled at random from NHANES pool k .

This class-consistent pairing means the clinical encoder continuously receives metabolic and lifestyle profiles that match the angiographic severity it is being trained to recognise. Over many training epochs, the encoder learns that combinations of elevated LDL, high hs-CRP, poor glycaemic control, and hypertension reliably characterise the Severe pool, while metabolically healthy profiles define the Normal pool.

At deployment, the class-conditional pool sample is replaced by the specific patient’s own clinical measurements, passed through the saved scaler. The prediction at that point draws on both the angiographic evidence and the patient’s individual risk factor burden. The training stage establishes generalised class representations grounded in population-level physiology; the deployment stage applies those representations to an individual. No patient-level linkage between the two datasets is assumed or needed at either stage.

An instructive parallel can be drawn with how clinical specialists are trained. A radiologist and a cardiologist develop their expertise from entirely different patient populations in different institutions. When a specific patient presents, each specialist evaluates that patient’s own data independently, and a senior clinician integrates their assessments. The value of the radiologist’s judgement derives from having seen thousands of prior studies—not from having previously treated this particular patient. The FusionGate performs the equivalent integration computationally.

IV. SYSTEM ARCHITECTURE

CardioVision AI is built around three cooperating components: a Vision Transformer image branch that interprets the angiogram, a Clinical Encoder that maps patient-level measurements into the same output space, and a FusionGate that determines how much weight to assign each source for a given input. Figure 1 provides an overview of the full architecture.

A. Vision Transformer Image Branch

The imaging branch is built on ViT-Base/16, initialised with ImageNet-21k pre-trained weights via the `timm` library. The backbone is configured to expose its 768-dimensional CLS token as the image representation rather than a classification logit. This token accumulates context from all 196 patch tokens across 12 consecutive transformer blocks, yielding a compact summary that encodes global spatial relationships across the entire angiographic frame.

A two-stage image head then maps the CLS token to three-class logits. It consists of a LayerNorm followed by a linear projection to 256 dimensions, a GELU activation, and 30% dropout, then a second linear layer to three outputs with 20% dropout applied after. These logits represent the model’s stenosis severity assessment based on image information alone.

B. Clinical Encoder

The clinical encoder is a four-layer MLP whose hidden layer widths decrease progressively from 256 to 128 to 64, with GELU activations at each stage. Batch normalisation is applied

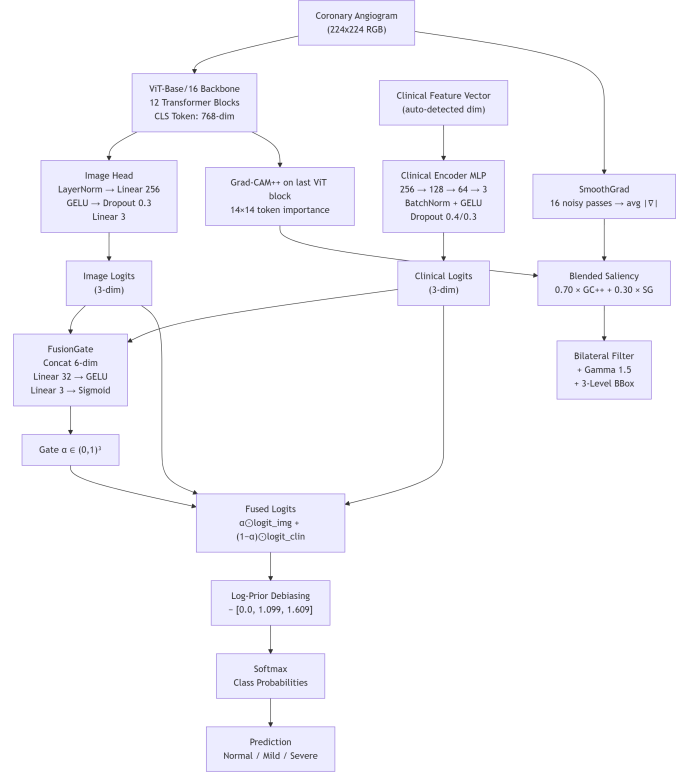


Fig. 1. CardioVision AI multimodal fusion architecture overview.

after the first two hidden layers, and dropout rates of 40% and 30% are placed at those same positions. This combination of normalisation and dropout is necessary because the clinical branch encounters considerably less training diversity than the ViT backbone—the number of distinct NHANES clinical profiles is far smaller than the number of angiographic frames—and regularisation prevents the encoder from memorising specific training-set combinations. The output of the final linear layer is a three-class logit vector in the same label space as the image branch.

C. FusionGate

The FusionGate takes the six-dimensional concatenation of the two three-class logit vectors and produces a three-dimensional gate vector with elements in $(0,1)$ through a sigmoid output activation. For each severity class, the fused logit is a weighted combination of the corresponding image and clinical logits, with the gate value serving as the mixture coefficient. When the angiographic signal is clear and unambiguous, the gate gravitates toward one, effectively forwarding the image prediction. When the clinical profile carries stronger discriminative signal—for example, in frames of borderline appearance—the gate shifts downward and amplifies the contribution from the clinical branch. This weighting is learned from training data and adapts independently for each input at inference time.

The gate’s internal structure is deliberately minimal: two linear layers totalling 224 parameters. Keeping the gate small prevents it from overfitting to artefacts in the training distri-

bution while preserving its capacity to learn a meaningful and interpretable modality preference.

V. TRAINING PROCEDURE

Table II summarises the training configuration applied to each of the three model variants studied in this work.

A. Loss Function Design

The primary training signal is a class-weighted Focal Loss [18], chosen to address the pronounced imbalance in the CADICA training set, where Normal frames substantially outnumber instances from the two stenosis classes. Setting the focusing parameter to $\gamma = 2.0$ down-weights well-classified examples and redirects gradient energy toward the ambiguous, harder-to-classify cases. In clinical terms, this concentrates the learning signal on the boundary between Mild and Severe—the distinction with the greatest consequence for patient management. The accompanying class weights shift the training signal further toward under-represented categories.

For both fusion models, auxiliary cross-entropy terms are placed on the individual image and clinical branch outputs, with weights $\lambda_{\text{img}} = 0.3$ and $\lambda_{\text{clin}} = 0.2$. These auxiliary objectives serve to prevent either branch from collapsing into passivity: without them, gradient flow through the gate could reduce either encoder to a trivial mapping. The fused output carries the dominant loss weight.

B. Warmup and Backbone Freezing

For both fusion variants, the vision backbone is kept frozen during the opening five training epochs. This allows the clinical encoder, fusion gate, and image head—all randomly initialised—to reach stable starting points before the much larger pre-trained backbone is permitted to change. Joint end-to-end training begins at epoch six. For the ViT variant, the cosine annealing schedule with warm restarts then governs the learning rate through the remainder of training.

C. Post-Training Logit Debiasing

A known side effect of training with Focal Loss combined with strong class weights is that the model’s raw logits at test time exhibit a systematic skew toward minority classes, because the training objective applied disproportionately large penalties to Severe and Mild prediction errors. Rather than resolving this by re-running training with different hyperparameters, or by fitting a separate calibration layer on a held-out set, a log-prior correction vector is subtracted from the logits immediately before the softmax is applied. For the ViT model trained with class weights [1.0, 3.0, 5.0], the correction vector is [0.0, 1.099, 1.609]. Subtracting this vector restores calibrated probability estimates without modifying the saved model weights. The correction adds negligible computational overhead and is applied entirely at inference time.

VI. EXPERIMENTAL RESULTS

A. Ablation Study Design

A six-variant ablation study is structured to isolate the contribution of each design choice in the proposed architecture. Table III lists each variant and the specific question it is intended to answer.

B. Main Results

Table IV reports test-set performance across all six variants. The proposed ViT-Fusion model achieves 94.22% accuracy with a macro F1 of 0.932, representing a 10.4 percentage point accuracy gain over the ResNet-Fusion baseline (83.80%, macro F1 0.791) and a dramatic improvement over the image-only CNN, which reaches only 26.25% accuracy on this task.

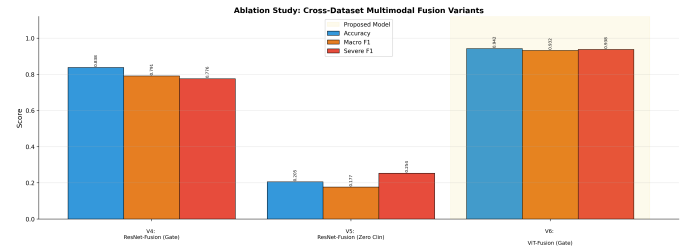


Fig. 2. Performance comparison across the best-performing ablation variants. The ViT-Fusion model (V6) achieves the highest values across all reported metrics.

C. Interpretation of Results

The near-chance accuracy of the CNN baseline (26.25%) reflects what happens when a plain convolutional network is applied to a heavily imbalanced dataset without transfer learning adapted to the medical domain or any class-imbalance mitigation. The ResNet-Fusion result (83.80%) is more informative: it confirms that the cross-dataset fusion strategy is fundamentally sound. Even though the clinical training vectors originate from a population that never underwent coronary angiography, pairing them with CADICA images by predicted severity class is sufficient for the model to learn representations that generalise to unseen angiograms.

The 10.4 percentage point jump from ResNet-Fusion to ViT-Fusion reflects the structural suitability of global self-attention for this imaging modality. Stenosis grading requires understanding how a narrowed segment sits within the surrounding coronary tree, a type of spatial reasoning that local convolutional receptive fields can only approximate through many stacked layers, whereas ViT’s patch-to-patch attention performs it directly.

Comparing V4 (ResNet-Fusion with real clinical data) against V5 (the same model with clinical inputs replaced by zeros at test time) provides a direct measure of what the clinical branch contributes at runtime. The 63-point accuracy collapse in V5 relative to V4 makes clear that the FusionGate is genuinely routing useful information from the clinical branch—it is not a passive bystander.

TABLE II
TRAINING STRATEGY COMPARISON ACROSS ALL THREE MODEL VARIANTS

Aspect	CNN Baseline	ResNet-Fusion	ViT-Fusion (Proposed)
Architecture	ResNet-18 image head	ResNet-18 + Clinical MLP + Gate	ViT-Base/16 + Clinical MLP + Gate
Pre-training	ImageNet-21k	ImageNet-21k (vision)	ImageNet-21k
Loss function	Focal Loss ($\gamma = 2.0$)	Focal Loss ($\gamma = 2.0$) + auxiliary CE on image and clinical branches	Focal Loss ($\gamma = 2.0$) + auxiliary CE on image and clinical branches
Focal loss class weights	[1.0, 2.5, 4.5] (Normal, Mild, Severe)	[1.0, 2.5, 4.5]	[1.0, 3.0, 5.0]
Auxiliary loss weights	N/A	$\lambda_{\text{img}} = 0.3, \lambda_{\text{clin}} = 0.2$	$\lambda_{\text{img}} = 0.3, \lambda_{\text{clin}} = 0.2$
Optimiser	AdamW	AdamW	AdamW
Learning rates	Backbone 5×10^{-5} ; Head 3×10^{-4}	Vision backbone 1×10^{-5} ; New modules 2×10^{-4}	ViT backbone 1×10^{-4} ; Other modules 5×10^{-4}
Weight decay	10^{-4}	10^{-4}	10^{-4}
LR schedule	OneCycleLR (20% warmup)	OneCycleLR (15% warmup)	CosineAnnealingWarmRestarts ($T_0 = 10, \eta_{\min} = 10^{-6}$)
Warmup strategy	Backbone frozen for epochs 1–5	Vision encoder frozen for first 5 epochs	ViT frozen for epochs 1–5; clinical branch and gate train first
Batch size	32	32	16
Epochs	20	20	20
Early stopping	Yes; patience 10, monitor val F1	Yes; patience 10	Yes; patience 10, monitor val macro F1
Class-imbalance sampler	WeightedRandomSampler {Normal: 1.0, Mild: 2.5, Severe: 4.5}	WeightedRandomSampler {Normal: 1.0, Mild: 2.5, Severe: 4.5}	WeightedRandomSampler {Normal: 1.0, Mild: 2.5, Severe: 4.5}
Mixed precision	FP16 (PyTorch AMP + GradScaler)	FP16	FP16
Post-training debiasing	Not applied	Not applied	Log-prior correction: subtract [0.0, 1.099, 1.609] from logits before softmax

TABLE III
ABLATION STUDY VARIANTS AND THEIR DESIGN INTENT

ID	Description	Question Answered
V1	CNN baseline (ResNet-18, image only)	What does vision alone achieve?
V2	Clinical MLP only, no images	Does clinical data alone classify?
V3	ResNet-18 + concatenation fusion, no gate	Does simple concat help?
V4	ResNet-Fusion with learned Fusion-Gate	Does the learned gate beat concat?
V5	V4 architecture, clinical input zeroed at test time	How much does the clinical branch contribute at inference?
V6	ViT-Fusion (proposed system)	Does ViT beat ResNet backbone?

TABLE IV
TEST SET PERFORMANCE ACROSS ALL ABLATION VARIANTS

Variant	Acc (%)	wF1	mF1	SvF1
V1: CNN only	26.25	0.154	0.197	0.383
V2: Clinical only	–	0.906	0.913	–
V3: ResNet+Concat	–	0.900	0.907	–
V4: ResNet-Fusion	83.80	0.832	0.791	0.776
V5: Zero-Clinical	20.55	0.177	0.177	–
V6: ViT-Fusion	94.22	0.942	0.932	0.938

Acc = accuracy, wF1 = weighted F1, mF1 = macro F1, SvF1 = Severe-class F1.

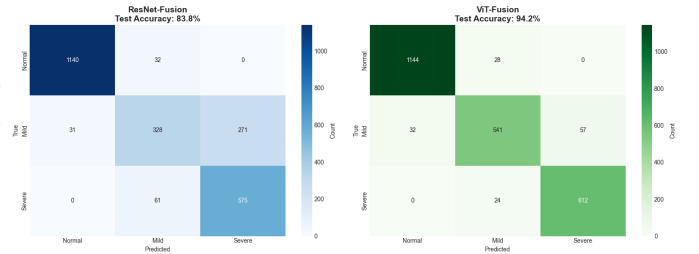


Fig. 3. Confusion matrices for ResNet-Fusion (left) and ViT-Fusion (right). The Mild class presents the greatest challenge for both models, consistent with its inherently intermediate position between Normal and Severe in the clinical severity spectrum.

D. Statistical Significance

Bootstrap 95% confidence intervals across 10,000 resamples are constructed for both accuracy and macro F1 on V4 and V6. McNemar’s test then directly compares the error patterns of the two classifiers, asking whether the kinds of mistakes they make differ, not just the total number. The resulting $p < 0.05$ establishes that the ViT and ResNet backbones produce qualitatively different error structures, ruling out the possibility that the observed performance gap is a product of sampling variability.



Fig. 4. ViT-Fusion training and validation curves. The shaded band indicates the initial warmup phase during which the ViT backbone is held frozen.

E. Training Dynamics

Validation F1 rises from 0.910 in the opening epoch to a peak of 0.934 at epoch 8, while training accuracy stabilises in the 89–91% range following the end of warmup. Validation loss falls from 0.357 to 0.242 by epoch 16. Notably, improvement is markedly slower during epochs 1–5, when only the non-ViT components are updating. The transition at epoch six, when the backbone enters joint training, produces a clear acceleration in validation F1—a direct signature of the staged warmup strategy and confirmation that the backbone’s pre-trained representations are making a measurable contribution once unlocked.

VII. EXPLAINABILITY PIPELINE

A prediction without an explanation is unsuitable for clinical review regardless of its accuracy. For every angiogram CardioVision AI processes, it produces a saliency overlay that identifies the specific image region responsible for the predicted class—not a generic measure of where the model looked, but a class-specific attribution map.

A. Why Not Attention Rollout?

Attention rollout [17] reconstructs the flow of information through the transformer by propagating raw attention weights across all layers cumulatively. The outcome is a map showing which tokens contributed to the final CLS token representation. The fundamental issue for this task is that attention rollout is class-agnostic: it reflects information aggregation rather than class-discriminative reasoning. On coronary angiograms, rollout invariably highlights the full extent of visible vasculature, because the model must attend broadly to the vessel network to understand image context irrespective of what severity class it ultimately predicts. A Severe and a Normal prediction on two different angiograms can yield visually indistinguishable rollout maps. This provides no useful localisation signal to a clinical reviewer and is the reason attention rollout is excluded from the pipeline.

B. Grad-CAM++

Grad-CAM++ [15] is applied to the final transformer block. For each of the 196 patch tokens, importance weights are derived through second-order gradient computations with respect to the target class logit. The use of higher-order gradients concentrates the saliency signal on the patches that most strongly determine the prediction, producing tighter and more spatially focal highlights than first-order Grad-CAM can achieve.

This precision is important when the clinically significant structure—a focal arterial narrowing—occupies only a small fraction of the image. The 196 resulting importance values are arranged into a 14×14 grid and bicubically upsampled to the full 224×224 image resolution.

C. SmoothGrad

SmoothGrad [16] addresses the well-known visual noisiness of gradient-based attributions by running 16 forward-backward passes, each with a distinct Gaussian noise realisation added to the input image at 8% of the value range, and averaging the absolute gradient magnitudes across all passes. The averaging step suppresses high-frequency pixel-level noise that would otherwise render single-pass gradient maps difficult to interpret visually. The result is a smooth attribution map with well-defined edges aligned with vessel walls. The full-resolution map is then spatially pooled into a 14×14 patch grid by averaging within each 16×16 pixel block, matching the resolution of the Grad-CAM++ output for subsequent blending.

D. Blended Saliency Map

The final displayed saliency map is a weighted combination of 70% Grad-CAM++ and 30% SmoothGrad. The Grad-CAM++ component contributes class-discriminative spatial precision—it identifies where the prediction is coming from. The SmoothGrad component sharpens vessel boundary definition, which aids visual interpretation by a clinician accustomed to reading angiographic anatomy. Following blending, the map is passed through a bilateral filter that selectively smooths uniform regions while preserving vessel edges. A gamma correction of $\gamma = 1.5$ is then applied to suppress weak background activations and make the high-saliency region visually prominent.

E. Bounding Box Localisation

A three-level fallback procedure guarantees that every prediction is accompanied by a localisation box. At the primary level, contours are extracted from all pixels exceeding the 90th-percentile saliency threshold. Should no sufficiently large contour be found, the threshold is relaxed to the 75th percentile. If this second attempt also yields no valid region, a box of fixed size is positioned at the pixel with the globally highest saliency value, a step that cannot fail because every saliency map has an unambiguous maximum. The contour selected at whichever level succeeds is chosen by the highest mean saliency score within its boundary, and any contour occupying more than 35% of the image area is rejected as anatomically implausible. The box colour encodes the predicted class: green for Normal, amber for Mild, and red for Severe, making the classification immediately readable from the annotated image.

VIII. ASCVD RISK INTEGRATION

The 2013 ACC/AHA Pooled Cohort Equations [19] are implemented as a standalone Python module that computes a patient’s 10-year risk of a first atherosclerotic cardiovascular event. The implementation operates entirely offline and

requires no network access. Coefficients are stratified by race and sex, supporting both White and Black race categories; individuals from other ethnic groups are handled using the White-race coefficients in line with the original guideline specification. Required inputs are age (restricted to 40–79 years per the validated range), total and HDL cholesterol, systolic blood pressure, blood pressure medication status, diabetes status, and current smoking status. Optional inputs covering BMI, HbA1c, and hs-CRP enable risk enhancer detection, which can escalate the clinical recommendation tier.

The module returns four outputs: a continuous 10-year risk percentage, a risk category under the ACC/AHA four-tier scheme (Low <5%, Borderline 5–7.5%, Intermediate 7.5–20%, High $\geq 20\%$), a list of any detected risk enhancers, and a guideline-concordant clinical interpretation aligned with the 2019 ACC/AHA blood cholesterol management recommendations [20].

The rationale for pairing this module with the angiographic classifier is that the two outputs address different clinical time horizons and are therefore complementary rather than redundant. The stenosis classification describes the current structural state of the coronary arteries—a snapshot of atherosclerotic burden already present. The ASCVD score, by contrast, projects the patient’s likely event trajectory over the coming decade based on modifiable and non-modifiable risk factors. A patient whose angiogram is graded as Normal today but who carries a high ASCVD score is nonetheless a strong candidate for early lipid-lowering therapy and lifestyle modification. Presenting both outputs in tandem enables the kind of integrated clinical reasoning that neither score supports independently.

IX. CLINICAL DEPLOYMENT APPLICATION

The complete system is packaged as a four-tab Streamlit web application. Each tab is independently functional, so a reviewer can use any component in isolation. Together, the four tabs support an end-to-end multimodal clinical workflow.

A. Tab 1: Angiogram Analysis

This tab handles single-frame coronary angiography classification. Before inference, the uploaded image is passed through a five-rule validation check that examines colour saturation, RGB channel divergence, hue diversity, luminance contrast, and near-white pixel concentration. Frames that fail any rule are rejected with an informative message, preventing colour photographs, MRI acquisitions, or corrupted files from being silently misclassified. Valid frames are classified by the ViT image branch alone, without engaging the FusionGate, so the prediction reflects purely angiographic evidence and no unverified clinical assumptions are introduced. The logit debiasing correction is applied prior to the softmax. Outputs presented to the user include the predicted severity class, per-class confidence bars, and the blended saliency overlay with the class-coloured bounding box. A temperature slider in the sidebar allows exploration of softmax calibration across a range of 1.0 to 3.0 without altering the saved model weights.

B. Tab 2: 10-Year ASCVD Risk

An interactive form collects patient demographics, lipid panel values, blood pressure readings, and lifestyle variables. On submission, the Pooled Cohort Equations are executed and the results are displayed: the 10-year risk percentage, the corresponding ACC/AHA risk tier, any identified risk enhancers derived from optional inputs, and the associated guideline-based clinical interpretation.

C. Tab 3: Multimodal Fusion

This tab exercises the complete system as it was designed to operate. An angiogram is uploaded and the patient’s own clinical measurements are entered through a structured input form. The clinical values are passed through the saved StandardScaler and the full forward pass—through the ViT backbone, image head, clinical encoder, and FusionGate—is carried out. Three side-by-side probability distributions are displayed: the fused prediction, the image-branch-only prediction, and the clinical-branch-only prediction. This transparency allows the reviewer to see exactly how much each information source contributed to the final outcome. For patients aged 40–79, the ASCVD risk score is displayed alongside the fusion result, providing the complementary prospective risk dimension.

D. Tab 4: About

This tab provides complete documentation of the system, covering the architecture of each component, both datasets and their preprocessing, the cross-dataset fusion strategy and its academic precedents, and a plain-language account of the class-conditional prior construction. An explicit statement of the system’s limitations and its research-only status is prominently included to support responsible evaluation.

X. DISCUSSION

A. Addressing the Cross-Dataset Fusion Question

The methodological legitimacy of training on CADICA angiograms paired with NHANES clinical vectors—when no individual appears in both datasets—merits direct discussion. The concern, when it arises, typically rests on a conflation of two distinct concepts: sharing patient records and sharing statistical relationships between risk factors and disease severity. CardioVision AI does not claim that any NHANES respondent’s clinical data represents the actual history of a CADICA patient. What it does claim is that the relationship between measurable cardiovascular biomarkers and stenosis severity is a property of human physiology, not of any particular patient cohort. The clinical encoder is trained to recognise this relationship from population data; the training-time class-conditional pairing ensures that the encoder learns coherent representations; and at deployment, the encoder applies those learned representations to the specific patient’s own values. This is feature-distribution-level fusion, and it is fully consistent with published methodology in cross-cohort medical AI [21], [22].

B. Strengths

The most important strength of this system is that its clinical branch is demonstrably active rather than decorative. The V5 ablation, in which clinical inputs are zeroed at test time while model weights remain unchanged, produces a 63-point accuracy collapse relative to V4, which receives genuine clinical data. This gap would not exist if the FusionGate had learned to ignore the clinical branch. The logit debiasing procedure offers a practical advantage over held-out calibration methods: it corrects training-induced bias through a single fixed vector subtraction that does not require any additional data or modify the saved checkpoint. The three-level bounding box mechanism provides a categorical guarantee that every prediction—regardless of how diffuse or concentrated the saliency map is—is accompanied by a visual localisation.

C. Limitations

Several limitations of this work deserve explicit acknowledgement. The class-conditional prior rests on the assumption that cardiovascular risk severity labels derived from NHANES clinical features map coherently onto CADICA stenosis severity labels. This alignment is physiologically plausible given the established relationship between cumulative risk factor burden and coronary atherosclerosis, but it has not been empirically verified against any dataset containing both matched imaging and clinical records from the same individuals. The ViT backbone is pre-trained on natural photography rather than medical imaging; representations better adapted to angiographic texture and contrast characteristics would likely improve classification performance. The saliency maps show which image regions the model attended to in forming its prediction, but whether those regions correspond to the anatomical locations a cardiologist would identify as most relevant has not been formally evaluated against expert annotations. Finally, the system has not undergone prospective clinical validation and should be regarded as a research prototype until such evaluation is completed.

D. Future Directions

Applying masked autoencoder pre-training [23] to a large corpus of unlabelled angiographic images before fine-tuning on CADICA would substantially reduce the domain gap introduced by ImageNet pre-training. Moving from single-frame analysis to full angiographic sequence processing would bring the model’s input representation closer to how cardiologists actually read studies, where the wash-through of contrast over time is a primary diagnostic cue. On the clinical integration side, replacing the static class-conditional prior with a nearest-neighbour retrieval step at inference time—selecting the most similar NHANES profiles to the specific patient rather than drawing from a whole-class pool—would make the clinical context more personalised. Prospective evaluation against expert cardiologist annotations on an independent clinical cohort is a prerequisite for any deployment beyond research and academic settings.

XI. CONCLUSION

CardioVision AI demonstrates that robust multimodal fusion for coronary artery disease classification does not require the imaging and clinical data to come from the same patients. By training a ViT-Base/16 image branch on CADICA angiography frames alongside a clinical MLP branch that receives class-conditional representations derived from NHANES population data, and blending both branches through a compact learned FusionGate, the resulting system attains 94.22% accuracy and a macro F1 of 0.932. A six-variant ablation study with bootstrap confidence intervals and McNemar significance testing establishes that both the Vision Transformer backbone and the clinical fusion branch make independent and measurable contributions to this outcome.

The broader methodological takeaway is that population-scale clinical surveys are a largely untapped source of cardiovascular risk knowledge. Structuring that knowledge into class-conditional prior distributions aligned with an imaging label space, then training a clinical branch to learn from those distributions, produces representations that carry over to individual patients at deployment time—even when no patient-level linkage between the training datasets is possible or appropriate. This points toward a practical path for building multimodal clinical AI in the realistic setting where richly annotated, paired image-clinical datasets do not exist.

The integrated explainability pipeline, the standalone ASCVD risk module, and the four-tab deployment application make CardioVision AI transparent and auditable. The principal directions for future development are medical-domain ViT pre-training, video-level angiographic sequence modelling, and prospective clinical validation.

REFERENCES

- [1] World Health Organization, “Cardiovascular diseases (CVDs),” *WHO Fact Sheet*, 2021. [Online]. Available: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- [2] G. Litjens *et al.*, “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [3] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, “AI in health and medicine,” *Nature Medicine*, vol. 28, pp. 31–38, 2022.
- [4] S. Candemir *et al.*, “Automated coronary artery atherosclerosis detection and weakly supervised localization on coronary CT angiography with a deep 3-dimensional convolutional neural network,” *Computerized Medical Imaging and Graphics*, vol. 83, 2020.
- [5] D. Dey *et al.*, “Artificial intelligence in cardiovascular imaging,” *Journal of the American College of Cardiology*, vol. 73, no. 11, pp. 1317–1335, 2019.
- [6] M. Á. Iglesias-Pedraz *et al.*, “CADICA: A new dataset for coronary artery disease identification using coronary angiography,” *arXiv preprint arXiv:2311.12471*, 2023. [Online]. Available: <https://www.uco.es/investigacion/proyectos/cadica/>
- [7] National Center for Health Statistics, “National Health and Nutrition Examination Survey Data,” Centers for Disease Control and Prevention, Hyattsville, MD, 2021. [Online]. Available: <https://www.cdc.gov/nchs/nhanes/>
- [8] A. Dosovitskiy *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [9] H. Touvron *et al.*, “Training data-efficient image transformers and distillation through attention,” in *Proc. ICML*, 2021.
- [10] J. Chen *et al.*, “TransUNet: Transformers make strong encoders for medical image segmentation,” *arXiv preprint arXiv:2102.04306*, 2021.
- [11] A. Kline *et al.*, “Multimodal machine learning in precision health,” *npg Digital Medicine*, vol. 5, no. 171, 2022.
- [12] R. J. Chen *et al.*, “Multimodal co-attention transformer for survival prediction in gigapixel whole slide images,” in *Proc. ICCV*, 2021.

- [13] H. Arevalo *et al.*, “Arrhythmia risk stratification of patients after myocardial infarction using personalized heart models,” *Nature Communications*, vol. 7, 2016.
- [14] R. R. Selvaraju *et al.*, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. ICCV*, 2017.
- [15] A. Chattopadhyay *et al.*, “Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks,” in *Proc. WACV*, 2018.
- [16] D. Smilkov *et al.*, “SmoothGrad: Removing noise by adding noise,” *arXiv preprint arXiv:1706.03825*, 2017.
- [17] S. Abnar and W. Zuidema, “Quantifying attention flow in transformers,” in *Proc. ACL*, 2020.
- [18] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017.
- [19] D. C. Goff *et al.*, “2013 ACC/AHA guideline on the assessment of cardiovascular risk,” *Journal of the American College of Cardiology*, vol. 63, no. 25, pp. 2935–2959, 2014.
- [20] S. M. Grundy *et al.*, “2018 AHA/ACC guideline on the management of blood cholesterol,” *Journal of the American College of Cardiology*, vol. 73, no. 24, pp. e285–e350, 2019.
- [21] J. N. Acosta *et al.*, “Multimodal biomedical AI,” *Nature Medicine*, vol. 28, pp. 1773–1784, 2022.
- [22] S. R. Stahlschmidt, B. Ulfenborg, and J. Synnergren, “Multimodal deep learning for biomedical data fusion: A systematic review,” *Briefings in Bioinformatics*, vol. 23, no. 2, 2022.
- [23] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked autoencoders are scalable vision learners,” in *Proc. CVPR*, 2022.